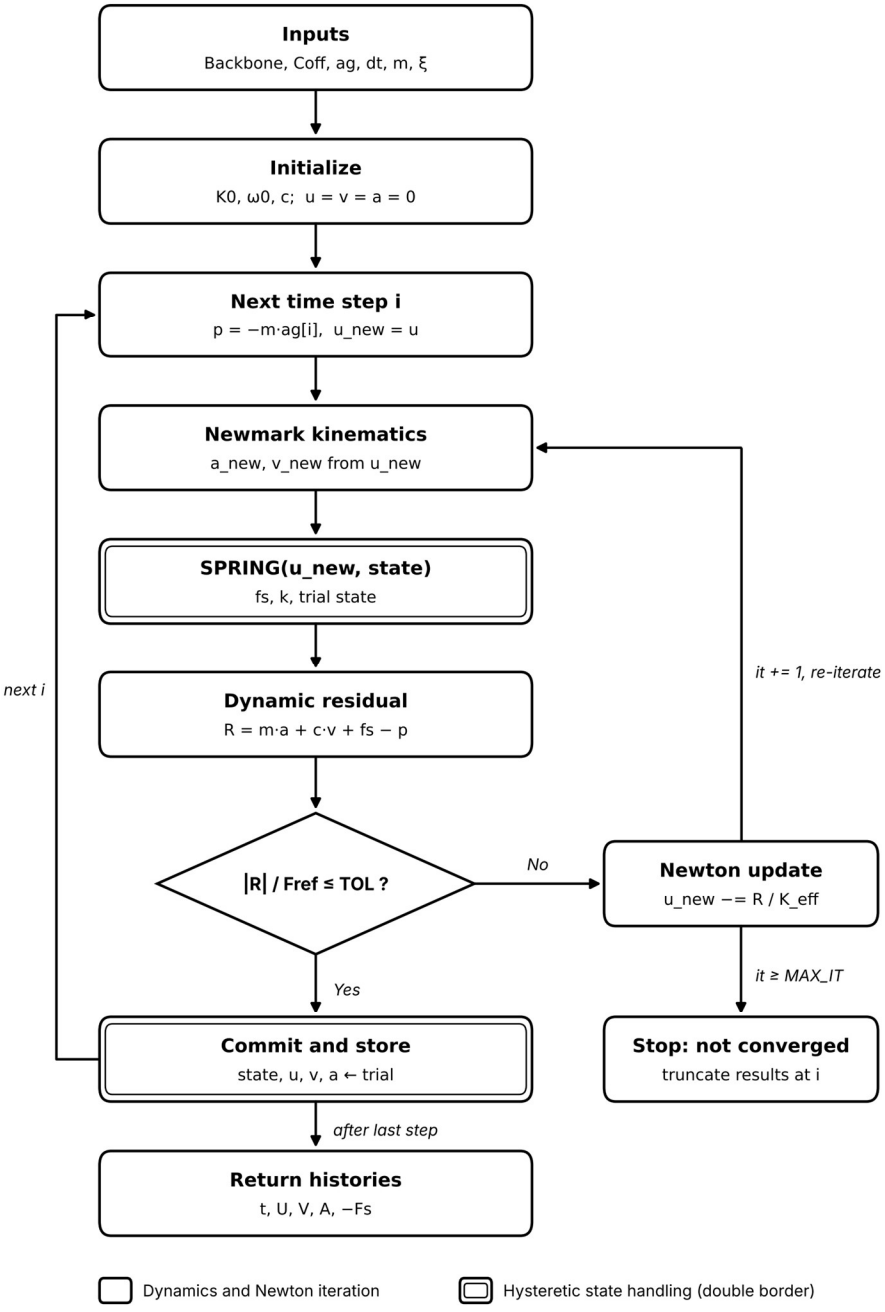
Flow Chart of the Numerical Method

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Nonlinear SDOF time-history: Newmark-β with Newton-Raphson and peak-oriented hysteresis



Step-by-step description

Step	Operation	Equation / action
1	Initialize	$K0 = \Sigma\, Coff \cdot FY/DY$; $\omega 0 = \sqrt{(K0/m)}$; $c = 2\cdot\xi\cdot\omega 0\cdot m$; $u = v = a = 0$
2	Load and guess	$p = -m\cdot ag[i]$; $u_{new} = u$ (previous committed displacement)
3	Newmark kinematics	$a_{new} = (u_{new} - u)/(\beta\cdot\Delta t^2) - v/(\beta\cdot\Delta t) - (1/(2\beta) - 1)\cdot a$; $v_{new} = v + \Delta t\cdot[(1-\gamma)\cdot a + \gamma\cdot a_{new}]$
4	Hysteretic spring	SPRING(u_new, committed state) returns fs, tangent k and trial state (state is not modified)
5	Residual	$R = m\cdot a_{new} + c\cdot v_{new} + fs - p$; converged if $\lvert R\rvert/Fref \le TOL$
6	Newton update	$K_{eff} = k + \gamma\cdot c/(\beta\cdot\Delta t) + m/(\beta\cdot\Delta t^2)$; $u_{new} \mathrel{-}= R / K_{eff}$; it += 1
7	Commit	state, u, v, a, fs ← trial values; store U, V, A, FS
8	Failure exit	If it ≥ MAX_ITERATIONS and not converged: print message, truncate results at step i

Key notes

- Equation of motion: $m\cdot u'' + c\cdot u' + Fs(u, history) = -m\cdot ag(t)$, with mass-proportional damping $c = 2\cdot\xi\cdot\omega 0\cdot m$.
- Default Newmark parameters are the average acceleration method ($\gamma = 0.5$, $\beta = 0.25$), which is unconditionally stable for linear systems.
- SPRING is a pure function of the committed state and the trial displacement, so it can be re-evaluated in every Newton iteration without corrupting the hysteresis history.
- State is committed only after the residual satisfies the tolerance, which matches the converged-step behaviour of OpenSees.
- The residual is normalized by $Fref = Csum \cdot \max(F)$, so the tolerance is dimensionless.
- Chart legend: single-border boxes are dynamics and Newton iteration; double-border boxes are hysteretic state handling.